



INTERNATIONAL JOURNAL OF CREATIVE RESEARCH THOUGHTS (IJCRT)

An International Open Access, Peer-reviewed, Refereed Journal

“Real time speech to Sign Language Translation”

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Abstract: The proposed system aims to support hearing-impaired individuals by translating real-time spoken English into American Sign Language (ASL) using a web-based solution. Built entirely with JavaScript, the system captures live speech using the SpeechRecognition Web API, processes the audio through AudioContext for extracting features like frequency, amplitude, and spectral information (including spectral centroid and bandwidth). It also utilizes Natural Language Processing (NLP) techniques via external Gemini AI APIs to refine and punctuate the transcribed text, and further analyze the emotional sentiment (positive, neutral, negative) of the spoken content. The processed content is then translated into ASL gestures, which are animated using Framer Motion and dynamically rendered through require.context (Webpack) for displaying sign images. Additional libraries such as Recharts are used for visualizing audio features and sentiment data, Axios for API communication, and Lucide-react for intuitive UI elements. This system aims to make real-time speech more accessible and inclusive, offering a meaningful, interactive communication experience for users with hearing or learning impairments.

Keywords - Hearing-impaired, real-time speech translation, American Sign Language, Speech Recognition, NLP, Sentiment Analysis, ASL Gestures, Emotion Detection.

I. INTRODUCTION

In a world where communication is a fundamental aspect of human interaction, it is essential to ensure that everyone has equal access to effective communication methods, regardless of any barriers they may face. One of the most prominent challenges faced by individuals with hearing impairments is the lack of accessibility to real-time spoken communication. While traditional sign language interpreters provide a valuable service, there is a growing need for innovative technologies to bridge the gap in communication, offering instant translation that can be easily accessed by individuals in real-time, without the need for human intermediaries.

The project, Real-Time Speech to Sign Language Translation, aims to address this need by creating an intelligent system that converts spoken English into American Sign Language (ASL) on a web-based platform. This system provides a seamless, accessible, and efficient method for hearing-impaired individuals to engage in real-time conversations, empowering them to interact with the world around them with ease and confidence.

The core functionality of the system revolves around the SpeechRecognition Web API, which captures live spoken English through a user's microphone. This speech is then processed and converted into text, which serves as the foundation for the translation. To ensure that the transcription is accurate and contextually meaningful, the system employs Natural Language Processing (NLP) techniques powered by advanced external AI APIs, such as Gemini AI. These APIs help refine and punctuate the transcribed text, making it more readable and accurate. In addition, the system integrates sentiment analysis to detect the emotional tone of the spoken content, categorizing it as positive, neutral, or negative, thus adding an emotional layer to the communication process.

Once the text has been processed and refined, it is translated into American Sign Language (ASL). This translation is represented through animated ASL gestures that are rendered and displayed dynamically on the web platform. To facilitate a smooth and engaging user experience, the gestures are animated using motion techniques, ensuring that they are fluid and visually representative of actual sign language movements. This process involves visualizing the gestures in real-time, offering hearing-impaired users a natural and intuitive communication interface.

The system's design incorporates additional features to enhance its usability and interactivity. For instance, audio features such as frequency, amplitude, and spectral information are analyzed to provide an insightful visualization of the sound's characteristics. This data is then displayed using interactive charts, allowing users to visually grasp the details of the spoken content.

The entire system is designed to be highly responsive and user-friendly, with intuitive UI elements ensuring a smooth and accessible experience for individuals with diverse needs.

The development of this real-time translation system is aimed at improving the daily lives of individuals with hearing impairments by providing them with a tool that enables them to access and participate in verbal communications effortlessly. By leveraging cutting-edge web technologies and AI, the system not only makes speech more accessible but also promotes inclusivity, allowing hearing-impaired individuals to engage in more meaningful, real-time interactions with their communities.

Furthermore, this system represents a significant step towards creating more inclusive digital environments. With the growing importance of web-based applications and platforms in everyday life, ensuring accessibility for all users, regardless of their hearing capabilities, is crucial. Through this project, the goal is to empower hearing-impaired individuals, enabling them to actively participate in both social and professional settings with the support of real-time speech-to-sign language translation technology. The project not only seeks to break down communication barriers but also aims to inspire further innovation in the field of accessibility and assistive technologies, creating a more inclusive and connected world for everyone.

- **Need of the work:** Sign language plays a vital role in helping individuals with hearing or speech impairments communicate in their daily lives. Among all sign languages, American Sign Language (ASL) has seen significant development, with many systems and tools already available. However, most of these systems are either limited in functionality or not real-time. There is still a need for accessible, interactive platforms that can translate spoken English into ASL gestures, especially in a web-based environment. This project addresses that gap by using modern web technologies to offer real-time speech-to-ASL translation.

- **Existing System:** Existing sign language translation systems commonly follow approaches like Direct Translation, Statistical Machine Translation, or Transfer-based Architecture. Most ASL systems focus on text-to-sign video conversion or avatar animations. However, few support real-time speech input, especially within browsers. Many of them also require complex installations or are restricted to desktop applications. Our work stands out by providing a lightweight, browser-based system that translates spoken English into ASL using live speech recognition and dynamic sign rendering.

- **Proposed System:** The proposed system is designed for individuals with hearing impairments who use ASL as their primary language. It addresses the communication barrier between hearing individuals and the deaf community by providing a real-time speech-to-ASL translation platform. Developed using JavaScript and modern front-end libraries, the system captures spoken English through the SpeechRecognition Web API, processes it using NLP techniques with support from external Gemini AI APIs for punctuation correction and sentiment analysis, and then translates the content into ASL gestures. These gestures are animated using Framer Motion and displayed using dynamically loaded sign images via Webpack's require.context. It also performs audio feature analysis—extracting frequency, amplitude, and spectral characteristics—visualized through Recharts graphs. This system is valuable for ASL learners, educators, or anyone aiming to improve communication with the ASL community. By combining accessibility, usability, and emotional context detection, the system enables more inclusive and meaningful interactions.

II. LITERATURE SURVEY

American Sign Language (ASL) has increasingly become a focal point in the development of assistive technologies aimed at enhancing communication for the deaf and hard-of-hearing community. One of the notable advancements in this domain is *DeepASL*, a system that leverages infrared sensing to non-intrusively capture ASL gestures without the need for wearable or camera-based systems. Developed with the goal of enabling both word-level and sentence-level translation, *DeepASL* employs a combination of a hierarchical bidirectional recurrent neural network (HB-RNN) and a Connectionist Temporal Classification (CTC) framework. The HB-RNN component enables the system to understand temporal dependencies in sign sequences from both forward and backward directions, while the CTC layer handles variable-length input and output sequences efficiently, allowing for smooth and real-time decoding. Evaluation results demonstrated an impressive word-level accuracy of 94.5%, along with an 8.2% word error rate when tested on unseen sentence inputs. These outcomes highlight the system's strong potential for real-world implementation in settings such as classrooms, public services, and everyday communication [1].

While vision-based systems like *DeepASL* offer significant promise, research has also extended into the realm of wearable technologies to explore more mobile and user-friendly ASL translation methods. Among the most prominent wearable solutions is the *Myo armband*, which is equipped with electromyography (EMG) sensors and inertial measurement units (IMUs). The EMG sensors detect muscle activity generated during sign gestures, while the IMUs capture movement, orientation, and acceleration data. In one key study, researchers utilized this armband to recognize ASL gestures by training a machine learning classifier, specifically a Support Vector Machine (SVM), on the captured biosignals. The combination of EMG features and motion data enabled the system to classify various ASL signs with high accuracy. This approach provided several benefits over traditional vision-based systems, including portability, better privacy, and the ability to function in poor lighting conditions. As a result, the *Myo armband* has gained attention as a viable tool for real-time and on-the-go sign language translation [2].

Expanding on the capabilities of single-sensor wearable devices, further research has focused on developing multimodal systems by integrating the *Myo armband* with other sensing technologies. One notable approach involves combining the armband with the Leap Motion controller—a device capable of capturing precise spatial and skeletal hand movement data. This fusion of EMG signals and spatial tracking significantly enhances the system's ability to interpret complex and nuanced gestures, particularly those that require detailed hand orientation and finger articulation. The integration of multimodal data leads to improved accuracy, robustness, and adaptability in various environmental conditions, such as crowded spaces or low-light scenarios. Moreover, these systems have been tailored to deliver feedback in both audio and visual formats, making them highly suitable for user-centric applications in education, healthcare, and personal communication. By emphasizing cost-effectiveness and mobile accessibility, such solutions have laid the groundwork for scalable and inclusive assistive technologies [3].

Despite the notable progress in ASL recognition systems, several challenges continue to hinder the goal of achieving seamless, natural translation. One of the key hurdles lies in capturing the subtleties of ASL grammar, which includes facial expressions, body posture, and spatial references that go beyond hand gestures. Additionally, variability in signing styles across individuals—due to differences in speed, hand size, muscle strength, or regional sign variations—poses a major challenge for the generalization of existing models. Research efforts are now directed toward developing adaptive models that can learn personalized patterns over time and adjust to user-specific behaviors. Moreover, real-time systems must strike a balance between computational efficiency and recognition accuracy to ensure fluid and responsive interactions. The future of this field lies in the continued integration of advanced sensor technologies, robust machine learning architectures, and thoughtfully designed user interfaces. These innovations are expected to drive the evolution of more intelligent, accurate, and user-adaptive ASL translation systems [4].

III. MOTIVATION

The motivation behind developing a Real-Time Speech to Sign Language Translation system stems from the need to bridge the communication gap between hearing individuals and the deaf or hard-of-hearing community. In daily life, people with hearing impairments often face challenges in understanding spoken conversations due to the lack of accessible, real-time translation tools. This project aims to make verbal communication instantly accessible by converting spoken speech into animated sign language using modern web technologies and audio processing techniques. By integrating tools like the SpeechRecognition API, AudioContext, and real-time animations through JavaScript frameworks, the system offers a dynamic and user-friendly solution. The broader goal is to contribute to a more inclusive society where technology assists in making communication equitable, especially in public services, education, and healthcare. This project also reflects a strong personal drive to build socially impactful solutions using data science and engineering skills.

In addition to its functional role, the system also adds emotional depth by analyzing tone and sentiment, making the translation more natural and expressive. Such emotionally-aware features help make interactions feel more human and less robotic, improving user engagement. The project holds strong potential to evolve into a scalable solution for mobile platforms, digital classrooms, or public service counters. It encourages awareness and sensitivity toward the challenges faced by the deaf community, promoting empathy through design. Overall, this project not only showcases technical innovation but also embodies a commitment to social equity and accessible technology for all.

As technology continues to advance, integrating such inclusive tools becomes not just innovative but necessary. The project pushes the boundary of accessibility by making speech and emotions visually understandable in real time. It also provides an opportunity to combine multiple fields—audio signal processing, natural language understanding, and front-end development—into one meaningful solution. Through this work, we aim to demonstrate how technology can become a tool of empowerment rather than exclusion. Ultimately, the project reflects a vision where communication is a universal right, not a privilege.

IV. METHODOLOGY

1. Input Audio

The system captures real-time speech from the user through a microphone using the Web Speech API or a similar voice input tool.

2. Feature Extraction

Audio features such as pitch, frequency, and amplitude are extracted using the Web Audio API (AudioContext). These features help in understanding the nature of the speech (e.g., emphasis, stress, etc.).

3. Learning Environment

The extracted features and recognized text are passed into a structured environment where predefined mappings, models, and logic guide the translation process.

4. Language Model

A lightweight language model interprets the sentence structure and helps correct or complete partial speech data. This improves the contextual understanding of the sentence.

5. Speech Recognition

The live audio input is converted into text using real-time speech recognition (e.g., Web Speech API). The output is a readable sentence or phrase.

6. Recognized Word Processing

The recognized text is tokenized to extract important keywords that are directly mapped to sign language gestures or animations.

7. Show Related Animation

Based on the identified keywords, corresponding sign language animations (GIFs or vector motions) are loaded and displayed smoothly using libraries like framer-motion.

8. Sentiment Analysis

The full sentence is analyzed for overall sentiment (positive, negative, or neutral) using basic NLP sentiment models. This gives emotional context to the speech.

9. Emotion Detection

A deeper emotion detection process identifies specific emotions like happiness, sadness, anger, etc., and adjusts the animation tone (e.g., facial expressions or speed) accordingly for a more natural output.

V. SYSTEM DESIGN AND IMPLEMENTATION

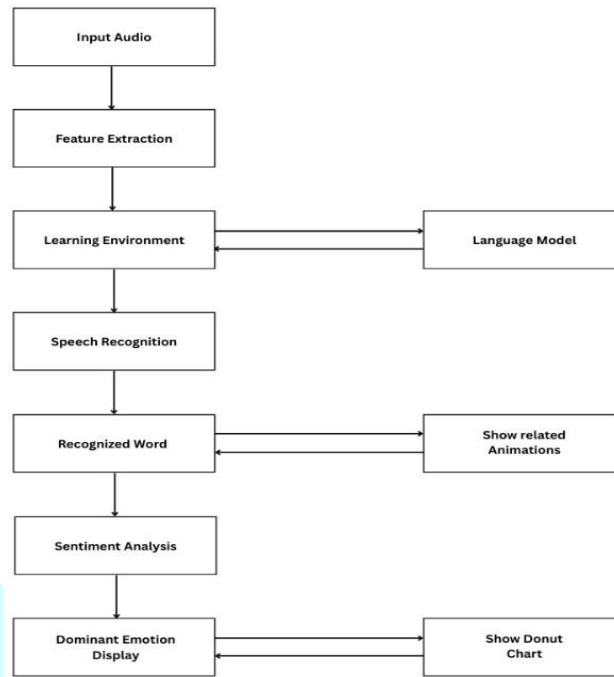


Fig a. System Architecture of Real Time Speech to Sign Language Translation

1. User Input Layer (Microphone Input)

- The user speaks into the device's microphone, which acts as the primary input to the system.
- The microphone captures real-time audio and sends it to both the speech recognition engine and the audio analysis module simultaneously.

2. Speech Recognition Module (Web Speech API)

- This module receives the audio input and converts it into text using the SpeechRecognition interface from the Web Speech API.
- It processes the speech in real-time, generating word-by-word transcripts.
- The output of this module is plain text, which is passed to the sign mapping module.

3. Audio Analysis Module (Web Audio API)

- In parallel, the audio input is processed by the AudioContext of the Web Audio API.
- This module extracts audio features like frequency, amplitude, and spectral patterns.
- These features are sent to the visualization module to generate real-time graphs.

4. Text-to-Sign Mapping Module

- This module takes the transcribed text and matches each word to a corresponding sign language image or symbol.
- It uses require.context() in JavaScript to dynamically load and retrieve image assets based on the spoken words.
- If a word has no available sign, a default icon or message is displayed.

5. Sign Rendering and Animation Module

- Once the sign images are selected, they are passed to this module, which handles the visual display.
- Using Framer Motion, it animates the transition between one sign and the next for a smooth visual experience.
- This helps mimic the natural flow of human sign language gestures.

6. Visualization Layer (Charts and Audio Feedback)

- The audio features from the analysis module are visualized using Recharts, a React-based charting library.
- Line charts or bar graphs display amplitude, frequency, and speech energy in real-time.
- This layer enhances interactivity and provides visual feedback to the user.

7. Frontend Interface (User Interface Layer)

- Built using React.js, this layer brings all components together into one cohesive interface.
- It includes:
 - A start/stop microphone button
 - A text display area for live transcription
 - A sign language display panel
 - An audio chart section
- It handles user interaction, real-time updates, and makes the application responsive across devices.

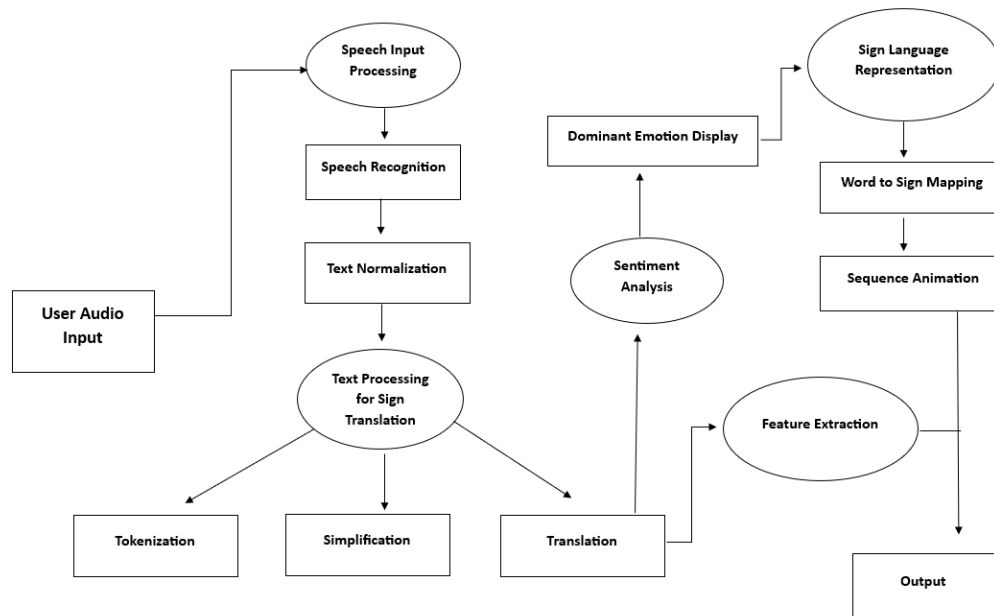


Fig b. Implementation Diagram

Implementation Details:

1. Speech Recognition Using Web Speech API

The project uses the **Web Speech API**, specifically the SpeechRecognition interface, to convert real-time spoken words into text. When the user speaks into the microphone, the API captures the audio and processes it to generate a text transcript. This happens continuously so the system can keep updating the spoken content. It handles voice input smoothly and works well in browsers like **Google Chrome** and **Microsoft Edge**. The recognized text is then sent to the sign mapping module for further processing.

2. Audio Feature Extraction with Web Audio API

In addition to speech-to-text conversion, the system also analyzes the audio signal using the **Web Audio API**, particularly through the AudioContext interface. This module helps extract low-level audio features such as:

- **Frequency** (pitch)
- **Amplitude** (volume)
- **Spectral energy** (distribution of energy over frequency)

These features provide insight into the nature of the spoken input. Although not directly used for translation, they are displayed as visual feedback to users for better interactivity.

3. Real-Time Visualization Using Recharts

The visual representation of audio features is handled using the **Recharts** library, a popular charting library for React. As the user speaks, different graphs such as line charts or bar graphs are updated in real-time to show:

- The waveform of the audio input
- Frequency variations
- Changes in amplitude

This feature makes the application visually informative and adds an analytical aspect for the user.

4. Mapping Speech to Sign Language Icons

Once the speech is converted into text, each word is matched with a corresponding **sign language symbol or image**. These signs are stored as static image files (e.g., PNG or SVG). The mapping is performed using JavaScript's `require.context()` function, which dynamically loads image files based on the spoken words. This allows the app to scale easily and manage a large number of sign images efficiently.

5. Smooth Animation with Framer Motion

To enhance the user experience, the transition between signs is animated using **Framer Motion**, a powerful animation library for React. When a new word is spoken, its corresponding sign image appears with smooth motion effects, such as fade-in or slide transitions. This makes the interaction more natural and engaging.

6. Clean Visual Elements with Lucide-react

Icons and visual cues in the application are displayed using **Lucide-react**, a modern icon library. These icons are used for UI components like the microphone button, audio status indicators, and user notifications. It keeps the interface visually clean and user-friendly.

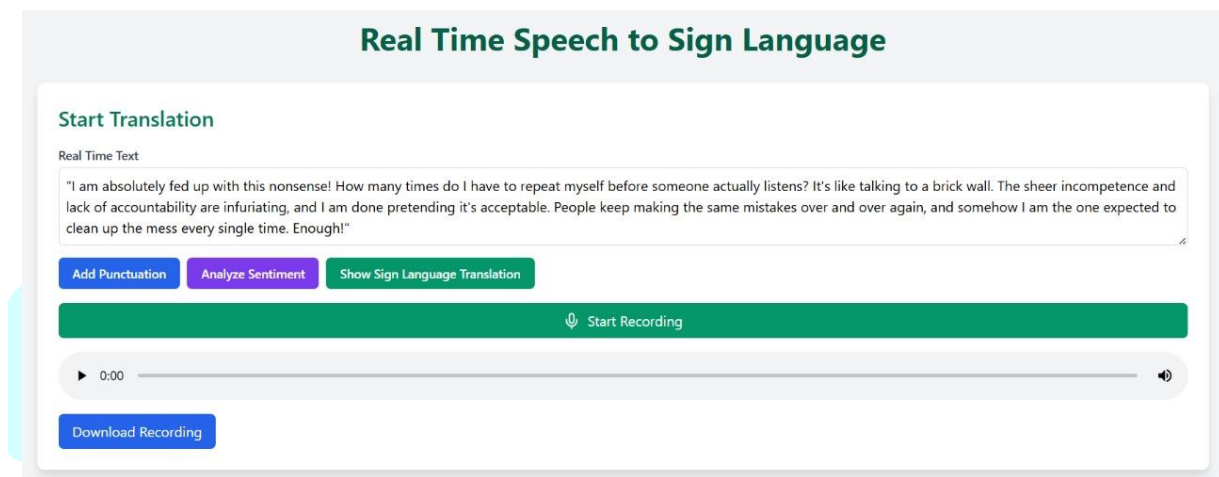
7. Frontend Development Using React.js

The entire user interface of the project is built using React.js, a JavaScript library for building component-based UIs. The interface includes:

- A microphone control button to start and stop speech recognition.
- A text area that displays real-time transcriptions of spoken words.
- A sign display panel that shows corresponding sign language images or icons.
- A visual chart area where audio features are plotted in real-time.

The application is fully responsive and can run on both desktop and mobile browsers, making it accessible to a wide range of users.

VI. RESULT ANALYSIS



This image showcases the input interface where users can provide audio input in two ways :

- The “Start Recording” button enables users to record their voice in real time.
- Alternatively, users can manually type the text into the "Real Time Text" input box.

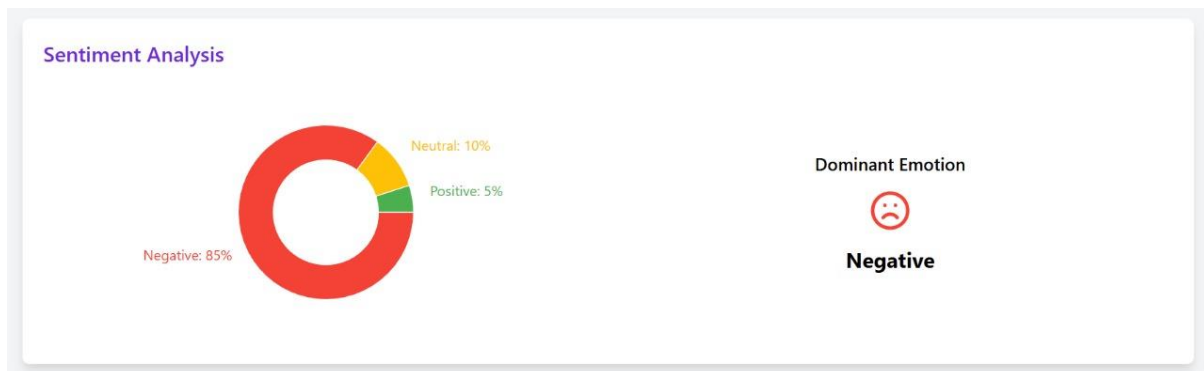
This dual-mode input flexibility ensures better accessibility and ease of use depending on the user's preference.

Additionally, three functional buttons have been added to enhance user control:

- “Add Punctuation” button processes the raw text using the Gemini API to add punctuation and capitalization.
- “Analyze Sentiment” button detects the emotional tone of the text and displays sentiment scores.
- “Show Sign Language Translation” button converts the processed text into animated ASL signs.

Users can also:

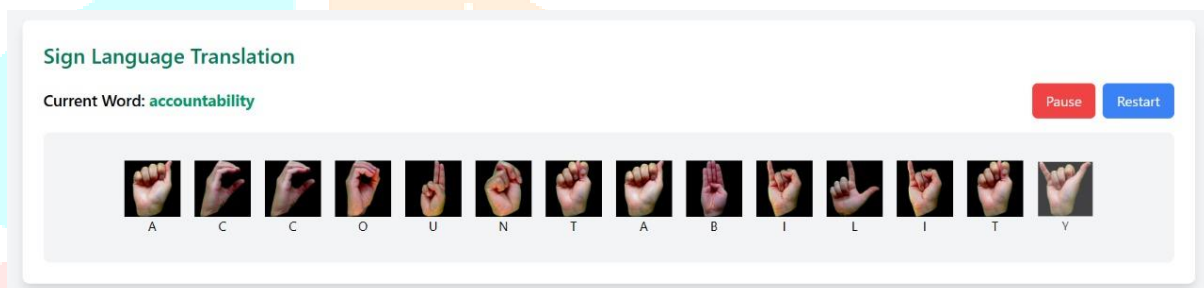
- Listen to the Recorded Voice by clicking the play button after recording.
- Download the Audio by using the provided “Download Recording” button to save the recorded file.



This image showcases the sentiment analysis features of the application :

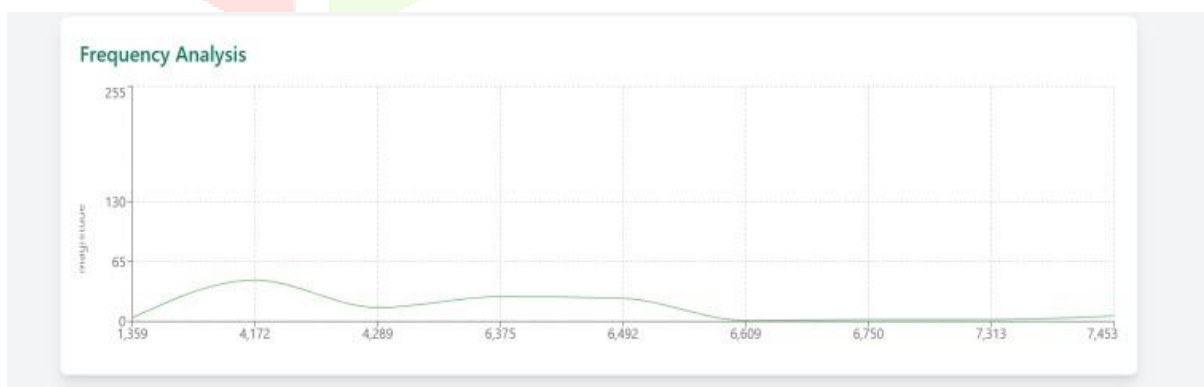
- The “Analyze Sentiment” button detects the emotional tone of the transcribed text and categorizes it into positive, negative, or neutral.
- On the left side, a donut chart visually represents the sentiment distribution with three categories :
 - Positive (green)
 - Negative (red)
 - Neutral (yellow)
- On the right side, the dominant emotion is displayed using a simple emoji icon, representing the overall emotion detected in the text.

This analysis provides users with a clear, visual representation of the emotional context of the speech.



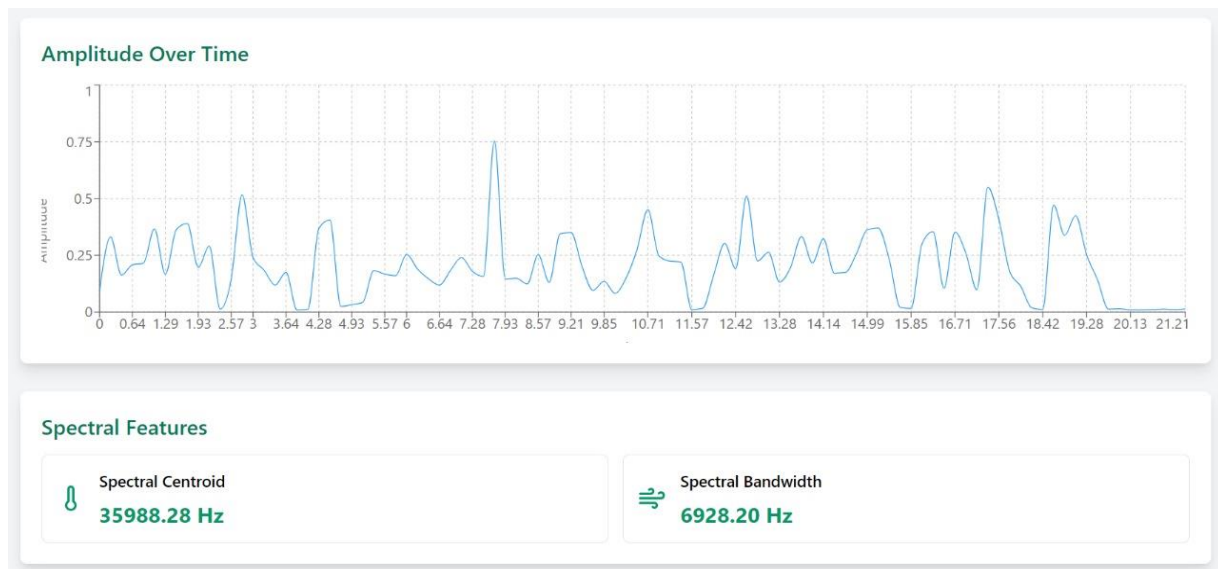
This image displays an animation of sign language gestures corresponding to the text entered or spoken:

- Each word from the input is translated into a visual sign language representation letter by letter.
- The animation shows one word at a time, allowing users to see the complete translation before moving to the next word.
- This mechanism helps users visually understand how each part of the text is signed, creating an intuitive connection between language and signs.



This image provides a detailed frequency spectrum analysis of the recorded voice.

- It shows how different frequencies are distributed in the audio signal.
- This analysis is crucial for understanding pitch, tone, and overall audio quality, and can be used in further signal processing or classification tasks.



This image presents both spectral features and time-based audio analysis.

- At the top, a waveform graph shows the amplitude over time, illustrating how loudness varies as the audio progresses.
- Below that, it extracts spectral centroid and spectral bandwidth, which are key features in analyzing the tone and clarity of the audio.

VII. CONCLUSION

The **Real-Time Speech to Sign Language Translation** project successfully demonstrates the potential of assistive technology in bridging communication gaps between the hearing and speech-impaired community and the rest of society. By converting spoken language into corresponding sign language gestures in real-time, this system offers an accessible and inclusive solution for everyday conversations.

1. Functionality and Efficiency

- The application effectively captures live speech using the Web Speech API and processes it using audio analysis techniques like frequency, amplitude, and spectral features.
- The visual feedback is delivered through intuitive animations and icons, built using React, Lucide-react, and Framer Motion, ensuring a responsive user interface.

2. Innovative Use of Frontend Technologies

- Tools like Recharts provided visual insights into speech patterns, enhancing user understanding and system transparency.
- The use of require.context and modular code structure allowed easy management of sign assets and real-time translation logic.

3. Impact and Social Relevance

- The system promotes inclusivity, accessibility, and empowerment for the speech- and hearing-impaired community.
- It can be expanded into various domains like education, customer service, healthcare, and public service announcements.

4. Scope for Future Work

- Integration of gesture generation using computer vision for full bidirectional communication (sign-to-speech).
- Support for multiple languages and dialects, making the system globally adaptable.
- Deployment as a mobile application for on-the-go accessibility.

VIII. REFERENCES

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